

Part A (or B, or C, or MSc)

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1. Let $Y \sim p(\cdot|\theta)$ with $Y = (Y_1, \dots, Y_n)$, $Y \in \mathcal{Y}$ and $\theta > 0$ a scalar parameter with prior $\pi(\theta)$.

(a) [15 marks] Suppose we observe $Y = y$ and wish to estimate θ .

- (i) Explain what it means for $\pi(\theta|y)$ to be *doubly intractable* and the consequence for MCMC-based estimation of θ given $Y = y$.
- (ii) Let $S(y) = (S_1(y), \dots, S_p(y))$ be a vector of summary statistics. For fixed $\rho > 0$ let $\Delta_y = \{y' \in \mathcal{Y} : |S(y) - S(y')| \leq \rho\}$. Let $\theta \in \Omega$ be discrete. Let $g(\theta)$ be a probability mass function and let $M = \max_{\theta} \pi(\theta)/g(\theta)$. Consider the following ABC algorithm.

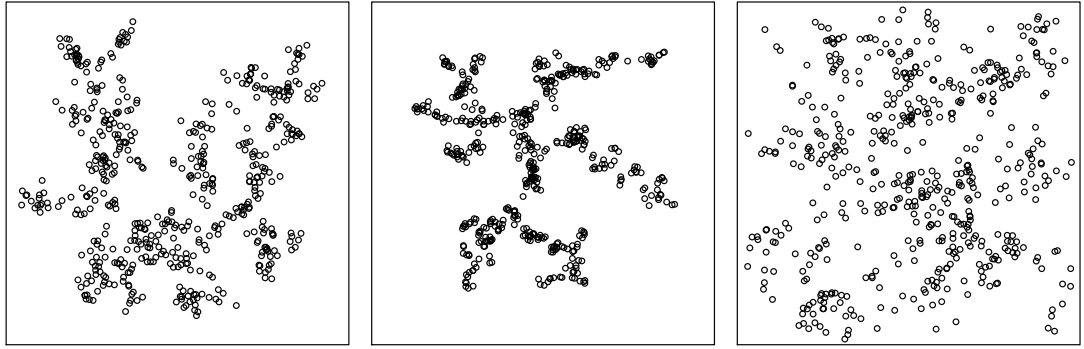
Step 1 Simulate $\theta \sim g(\theta)$, $y' \sim p(y'|\theta)$ and $u \sim U(0, 1)$.

Step 2 If $U \leq \frac{\pi(\theta)}{Mg(\theta)} \mathbb{I}_{y' \in \Delta_y}$ stop and return θ , otherwise goto Step 1.

Show that this algorithm targets $\pi_{ABC}(\theta|y) = \pi(\theta|Y \in \Delta_y)$.

- (iii) An ABC algorithm yields samples $\theta^{(t)}$, $t = 1, 2, \dots, T$ with associated statistics $S^{(t)} = (S_1^{(t)}, \dots, S_p^{(t)})$. Explain how to make a regression adjustment to the sampled θ -values. What is the purpose of this adjustment?

(b) [10 marks] Data on locations of large ancient Danish barrow-graves are plotted (left) in the scaled region $A = [0, 1]^2$. The following model generates n grave locations $y_1, \dots, y_n$.



For $i = 1, \dots, n$, the i th person dies at a location $x_i \sim \pi(\cdot|\psi)$ in A , with ψ a parameter controlling the distribution of x_i in A . The first person to die is buried where they died, at $y_1 = x_1$; the i 'th person is buried at $y_i \sim N(y_{j_i}, \sigma^2 I_2)$, where $j_i = \arg \min_{k \in 1, \dots, i-1} |x_i - y_k|$ is the nearest pre-existing grave and $\sigma > 0$ is unknown. Simulations in the figures above show (centre) $\sigma = 0.02$ and (right) $\sigma = 0.04$ with $n = 1000$ with $\pi(\cdot|\psi)$ uniform.

- (i) Write down the posterior for ψ , $x = (x_1, \dots, x_n)$ and σ , including any missing data needed to make the likelihood well defined.
- (ii) Assume priors and simulators for all unknown parameters, and for y given the parameters, are available. Explain how to generate samples ψ and σ from the ABC posterior. You will need to suggest suitable summary statistics and give an ABC algorithm as pseudo-code.

2. (a) [8 marks]

- (i) Write down the first three of the Savage axioms, in the form given by de Groot.
- (ii) For $y_i \in \{0, 1\}$ for $i = 1, 2, 3, \dots$, the sequence of distributions $\pi_n(y_1, \dots, y_n)$, $n = 1, 2, 3, \dots$ is *marginally consistent* if

$$\pi_n(y_1, \dots, y_n) = \pi_{n+1}(y_1, \dots, y_n, 0) + \pi_{n+1}(y_1, \dots, y_n, 1).$$

Suppose our prior for the events $Y_1 = y_1, \dots, Y_n = y_n$ is given by $\pi_n(y_1, \dots, y_n)$ and that these distributions are not marginally consistent. Show that our preferences on the events $Y_1 = y_1, \dots, Y_n = y_n$ do not satisfy the Savage axioms.

- (b) [9 marks] Consider the following model for a random binary $n \times n$ matrix G with diagonal entries all equal zero. For $i = 1, \dots, n$ let $z_i = (z_{i,1}, z_{i,2})$ be a point taken independently and uniformly at random in $[0, 1]^2$. Let $z = (z_1, \dots, z_n)$. The random matrix $G = G(z)$ is defined as follows. For each i, j in $1, \dots, n$ let

$$G_{i,j} = \begin{cases} 1 & \text{if both } z_{i,1} > z_{j,1} \text{ and } z_{i,2} > z_{j,2} \\ 0 & \text{otherwise.} \end{cases}$$

Let Ω_n be the set of all $n \times n$ binary matrices which can be generated in this way and let $\pi_n(g) = \Pr(G = g)$ for $g \in \Omega_n$.

- (i) For $G \in \Omega_n$ let $G^{(-n)}$ be obtained from G by deleting the last row and column. Show that if $G \sim \pi_n$ then $G^{(-n)} \sim \pi_{n-1}$ and briefly interpret.
 - (ii) Let $P_n(g) = 1/|\Omega_n|$ for $g \in \Omega_n$, so P_n is the uniform distribution. Show that if $G \sim P_n$ then $G^{(-n)}$ is not in general distributed according to P_{n-1} . *Hint: the set sizes are $|\Omega_2| = 3$, $|\Omega_3| = 19$ and $|\Omega_4| = 219$.*
 - (iii) Suppose that for each $n = 1, 2, 3, \dots$ our prior for $n \times n$ binary matrix G is P_n . Explain why these preferences over events do not satisfy the Savage axioms.
- (c) [8 marks] We can think of G as representing a directed graph with a directed edge from i to j if $G_{i,j} = 1$ and no directed edge from i to j if $G_{i,j} = 0$. Suppose G is unknown and our data are J random walks $(a_1, \dots, a_K)$ of length K on the nodes of G .
- 1. The first node $a_1 \sim U\{1, \dots, n\}$ is chosen at random.
 - 2. For $k = 1, \dots, K - 1$ choose $a_{k+1} \sim U\{j : G_{j,a_k} \neq 1\}$ at random from nodes j with no directed edge pointing back from j to a_k (a_k itself is always in this set).

Suppose we observe J walks $A_j = (a_{j,1}, \dots, a_{j,K})$, $j = 1, \dots, J$. Let $A = \{A_1, \dots, A_J\}$ denote the set of observed walks.

- (i) Write down the posterior $\pi(z|A)$ in terms of $G(z)$ and A for the prior $G \sim \pi_n$.
- (ii) Explain how you would use MCMC to sample $\pi(G|A)$. Give an explicit algorithm. There is no need to simplify acceptance ratios or check irreducibility.
- (iii) Explain how you would use your MCMC to estimate the posterior probability that all entries in G are zero, and how you might estimate the unknown true G itself.

3. Let $M > 0$ be a positive integer and let $\alpha > 0$ be a real positive scalar. Let $w = (w_1, \dots, w_M)$. Consider the following model,

$$\begin{aligned} w &\sim \text{Dirichlet}(\alpha_1, \dots, \alpha_M) \quad \text{with } \alpha_m = \alpha/M, m = 1, \dots, M \\ z_i|w &\sim \text{Multinom}(w), i = 1, \dots, n. \end{aligned}$$

Here for $i = 1, \dots, n$, $z_i \sim \text{Multinom}(w)$ means $z_i = m, m \in \{1, \dots, M\}$ with probability w_m .

- (a) [16 marks] Suppose the list $z_1, \dots, z_n$ of cluster labels contains $K \leq M$ unique values $m_1, \dots, m_K$. For $k = 1, \dots, K$ let $C_k = \{i : z_i = m_k, i = 1, \dots, n\}$ and $C = \{C_1, \dots, C_K\}$.
 (i) Show that in this model the probability to generate partition C is

$$P(C) = \frac{\Gamma(\alpha)}{\Gamma(\alpha/M)^K} \frac{M!}{(M-K)!} \frac{\prod_{k=1}^K \Gamma(\alpha/M + |C_k|)}{\Gamma(\alpha + n)}$$

where $|C_k|$ is the number of elements in C_k for $k = 1, \dots, K$.

Hint: The Dirichlet probability density for the probability vector w is

$$\text{Dirichlet}(w; \alpha_1, \dots, \alpha_M) = \frac{\Gamma(\sum_{m=1}^M \alpha_m)}{\prod_{m=1}^M \Gamma(\alpha_m)} \prod_{m=1}^M w_m^{\alpha_m - 1}$$

- (ii) Let $C_{-i} = \{C_{-i,1}, \dots, C_{-i,K-i}\}$ be the partition we obtain by removing i from its set in C . The partition C_{-i} contains K_{-i} sets $C_{-i,k}, k = 1, \dots, K_{-i}$ following the removal of i . Here K_{-i} may be less than K since if i is in a set by itself we remove that set entirely.

Suppose we observe C_{-i} and we consider the distribution of $C|C_{-i}$. Show that

$$P(i \in C_k | C_{-i}) \propto \begin{cases} \frac{\alpha}{M} + |C_{-i,k}| & \text{if } k \leq K_{-i} \text{ and} \\ \frac{\alpha}{M}(M - K_{-i}) & \text{if } k = K_{-i} + 1. \end{cases}$$

Hint: the Gamma function satisfies $x\Gamma(x) = \Gamma(x+1)$.

- (iii) Find a generalisation of the Chinese restaurant process generating random partitions C of n elements according to $P(C)$.
 (b) [9 marks] Now consider a mixture model for data $y = (y_1, \dots, y_n)$. Suppose $C \sim P(C)$ is a random partition of $1, \dots, n$ into K sets. The cluster to which y_i belongs is $z_i = \{k : i \in C_k\}$. Each cluster has an observation model parameter θ_k^* distributed as

$$\theta_k^* | C \sim H, k = 1, \dots, K$$

independent and identically distributed according to H , where H is a distribution on $\mathcal{R}$ with probability density h say. Let $\theta^* = (\theta_1^*, \dots, \theta_K^*)$. The observation model is

$$y_i | C, \theta^* \sim f(y_i | \theta_{z_i}^*), i = 1, \dots, n,$$

independent given C and θ^* , with f the density of a real scalar variable and $y \in \mathcal{R}^n$.

- (i) Write down the posterior $\pi(C, \theta^* | y)$ for C, θ^* given y .
 (ii) Give a Gibbs sampler updating C, θ^* to D, ψ^* which takes each $i = 1, \dots, n$ in turn and assigns it a new cluster by sampling the conditional distribution for the new cluster assignment $P(i \in D_k | C_{-i}, \theta_{-i}^*, y)$, explaining how ψ^* is determined.